Distributed Finite-Time Consensus Tracking Control for Nonlinear Multi-Agent Systems With FDI Attacks and Application to Single-Link Robots

Yuqiang Jiang, Ben Niu, Xinjun Wang, Xudong Zhao, Huanqing Wang, and Bocheng Yan

***Abstract*—This brief considers the distributed finite-time consensus tracking control problem for nonlinear multi-agent systems with false data injection (FDI) attacks. Firstly, since instability is inevitable when false datas are injected into the researched system, a security control algorithm combining the modified coordinate transformation and the dynamic surface control (DSC) method is developed to eliminate the negative impact of FDI attacks and reduce the burden caused by iterative computations of virtual controls. Besides, there are unknown nonlinearities in the dynamics of each agent, which can not be directly handled by the traditional control design. Thus, the neural-network-based approximation technique is introduced to compensate the unknown nonlinearities. In summary, a novel controller is constructed for each agent, which guarantees the consensus of the outputs of all agents can be achieved and that all the signals of the studied system are bounded within a finite-time period. To prove the correctness of the designed security control algorithm, a simulation example of the multi-agent system composed of multiple single-link robots is presented.**

***Index Terms*—Consensus tracking control, finite-time control, false data injection attacks, nonlinear multi-agent systems, single-link robots.**

I. Introduction

DURING the preceding thirty years, since the distributed control of multi-agent systems contains research potential greatly, some results from different aspects have been obtained, such as consensus [1], formation [2]. Specifically, consensus control is more general than other research topics on multi-agent systems [3]. Consensus signifies that the outputs (or states) of followers and leaders tend to the same value by using information from their neighbors. It is remarkable that the consensus tracking control is the key research topic because it is more resource efficient than the leaderless consensus control. Moreover, with the rise of intelligent technology, the consensus tracking control of nonlinear multi-agent systems has been further developed and some excellent intelligent control algorithms have been proposed [3], [4]. In [3], [4], the control algorithms based on the neural networks (NNs) approximation technology not only approximate the unknown nonlinearities of each agent and its neighbors but also guarantee the bounded stability of the researched systems. It is pointed out that the convergence or boundedness of many engineering systems usually requires to be met in a finite-time period. Compare with the bounded stability, the finite-time stability has stronger anti-jamming ability and rapider convergence to uncertainties [5]. Thus, how to ensure that the nonlinear multi-agent system reaches stability in finite time is worth studying.

The finite-time stability theory is first proposed by Weiss and Infante in 1960s in [6] and then some finite-time control schemas for the nonlinear multi-agent systems have been obtained in the past few years [7], [8]. Due to the complicity of network environment, the engineering network systems face the risk of malicious attacks. However, the security problem is not enough considered in the existing finite-time stability results [7], [8], this prompt us to research the finite-time control problem in an insecure network.

As more and more control systems are deployed in complicated network, the security control problem of network systems is the subject of intense research today. Common malicious attacks include FDI attacks, replay attacks and denial-of-service attacks. Among them, the FDI attacks are more general way of network attacks and have strong destructive ability, which mean the attackers inject the false datas into a system to change the states of the system. Considering the destructiveness of the FDI attacks, many scholars have paid more and more attention to the research on the FDI attacks [9], [10]. However, the existing results are only appropriate to the decentralized control of cyber-physical systems. It is still a challenge to overcome the difficulties of distributed security control of nonlinear multi-agent systems under the FDI attacks.

This brief addresses the finite-time consensus tracking control problem of the nonlinear multi-agent systems with the FDI attacks. The main contributions of this brief can be stated as follow. ***(i)*** Unlike the existing ideas of dealing with the FDI attacks [11], [12], where the proposed control strategies only

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considered the single system and the tracking control problem under the FDI attacks has not been solved, a security control strategy is proposed is this brief based on the modified coordinate transformation to solve the distributed consensus tracking control problem under the FDI attacks. ***(ii)*** Compared with the previous decentralized control results [13], [14] that design the decentralized controller, this brief proposes a distributed controller for each agent, which ensures the consensus of the outputs of all agents and the boundedness of all closed-loop signals within a finite-time period. ***(iii)*** Different from the existing results [11], [12], [15], the more practical situations are considered in this brief, where different system states of the one follower suffer from the different attack signals.

II. Preliminaries and Problem Formulation

A. Graph Knowledge

Let $G = (E, N_d)$ represents the direct topology graph of N agents, where $E = \{e_1, \ldots, e_N\}$ is a group of nodes and $N_d \subseteq E \times E$ is a group of edges. The edge $(e_i, e_j) \in E$ denotes that the ith agent can receive the datas from its neighboring jth agent. The followers' adjacency matrix is chosen as $A = [a_{i,j}] \in \mathbb{R}^{N\times N}$, which represents the relationship of followers in the graph G. When the ith agent can receive the datas from its neighbors, then $a_{i,j} > 0$, and otherwise $a_{i,j} = 0$. The followers' in-degree matrix is selected as $D = diag\{\sum_{j=1}^{N} a_{1,j}, \ldots, \sum_{j=1}^{N} a_{N,j}\}$. The followers' Laplacian matrix is chosen as $L = D - A$. The graph G only contains one leader, which only can send datas to its neighboring followers. Thus, the leader's adjacency matrix is chosen as $B = diag\{b_{1,0}, \ldots, b_{N,0}\}$.

B. System Descriptions and Assumptions

The ith agent system model is described by

$$\begin{cases} \dot{x}_{i,m} = g_{i,m}(\bar{x}_{i,m})x_{i,m+1} + f_{i,m}(\bar{x}_{i,m}) + \varpi_{i,m}, \\ \dot{x}_{i,n} = g_{i,n}(\bar{x}_i)u_i + f_{i,n}(\bar{x}_i) + \varpi_{i,n}, \\ y_i = x_{i,1}, 1 \le m \le n-1, \end{cases} \tag{1}$$

with

$$\breve{x}_{i,l}(t) = x_{i,l}(t) + \beta_{i,l}(x_{i,l}, t), l = 1, \ldots, n, \tag{2}$$

where $i = 1, 2, \ldots, N$, $\bar{x}_{i,m} = (x_{i,1}, \ldots, x_{i,m})^T$, $\bar{x}_i = \bar{x}_{i,n}$, $u_i \in \mathbb{R}$ and $y_i \in \mathbb{R}$ are the state vectors, control input and control output, respectively. $f(\cdot)$, $g(\cdot)$ are the unknown smooth nonlinearities, $\varpi_{i,l}$, $l = 1, \ldots, n$, represent the unknown disturbances. It is assumed that the attacks are state-dependent and are parameterized as $\beta_{i,l}(x_{i,l}, t) = \gamma_{i,l}(t)x_{i,l}(t)$ with unknown time-varying signals $\gamma_{i,l}(t)$. Further, the post-attack system states are reconstructed as $\breve{x}_{i,l} = \rho_{i,l}(t)x_{i,l}$, where $\rho_{i,l}(t) = 1 + \gamma_{i,l}(t)$, $l = 1, \ldots, n$.

The leader can be modeled as

$$\begin{cases} \dot{x}_r = f_r(x_r, t), \\ y_r = x_r, \end{cases} \tag{3}$$

with

$$\breve{y}_{i,r} = \rho_{i,1}(t)y_r, \tag{4}$$

where the continuous function $f_r(x_r, t)$ satisfies locally Lipschitz condition for $t \ge 0$, y_r represents the output of dynamics and $\breve{y}_{i,r}$ is a actual measurable output after the communication link from the leader to the ith follower is attacked, $i \in N$.

Remark 1: Using the FDI attacks model (2), the case where the output of leader is injected with the same false data as the neighbor followers is considered.

Assumption 1 [12]: The signals of $\gamma_{i,l}(t)$ are positive and satisfy $\underline{\gamma}_{i,l} \le \gamma_{i,l}(t) \le \bar{\gamma}_{i,l}$, where $l = 1, \ldots, n$, $\underline{\gamma}_{i,l}$ and $\bar{\gamma}_{i,l}$ are the unknown constants.

Remark 2: Assumption 1 is usually adopted as a general assumption for the FDI attacks and is widely used in the existing works [11], [12].

Assumption 2 [16]: For any given the constant $\bar{x}_r > 0$ and the continuous function $f(\cdot)$, the subsequent inequalities $|x_r(t)| \le \bar{x}_r$ and $|f_r(x_r, t)| < f(x_r)$ can be obtained.

Assumption 3 There are unknown parameters $\underline{g}_i^*$ and $\bar{g}_i^*$ such that the function $g(*)$ with known sign satisfies

$$0 < \underline{g}_i^* \le |g_i(*)| \le \bar{g}_i^*. \tag{5}$$

Lemma 1 [17]: If exist some design constants a, $b > 0$, $0 < c < 1$, $0 < p < \infty$, such that the finite-time stable Lyapunov function satisfies

$$\dot{V}(x) \le -aV(x) - bV^c(x) + p. \tag{6}$$

Furthermore, the set of residuals of the system solution is calculated as

$$\lim_{t\to T} V(x) \le \min\{\frac{p}{(1-\mu)a}, [\frac{p}{(1-\mu)b}]^c\}, \tag{7}$$

where μ satisfies $0 < \mu < 1$, and the setting time satisfies

$$T \le \max\left\{t_0 + \frac{1}{\mu a(1-p)} \ln \frac{\mu a V^{1-p}(t_0) + b}{b}, \right. \\ \left. t_0 + \frac{1}{a(1-p)} \ln \frac{aV^{1-p}(t_0) + \mu b}{\mu b}\right\}. \tag{8}$$

Lemma 2: For $x_{i,l} \in \mathbb{R}$, $l = 1, \ldots, n$, $0 \le k \le 1$, we can get

$$\left\{\sum_{m=1}^{n} |x_{i,m}|\right\}^k \le \sum_{m=1}^{n} |x_{i,m}|^k. \tag{9}$$

Lemma 3: For $m_1 > 0$, $m_2 > 0$, $m_3 > 0$, $\lambda_1 \ge 0$, $\lambda_2 \ge 0$, $\lambda_3 \ge 0$, one has

$$\lambda_1^{m_1}\lambda_2^{m_2}\lambda_3 \le m_3\lambda_1^{m_1+m_2} + \frac{m_2}{m_1+m_2}\left[\frac{m_1}{m_3(m_1+m_2)}\right]^{\frac{m_1}{m_2}} \lambda_2^{m_1+m_2}\lambda_3^{\frac{m_1+m_2}{m_2}}. \tag{10}$$

III. Main Result

For the researched system (1), the backstepping-based finite-time security control strategy contains n steps. Considering that only the post-attack output information of neighbors is obtained for the ith agent. The modified coordinate transformation is constructed as

$$z_{i,1} = \sum_{j=1}^{N_i} a_{i,j}(\breve{y}_i - \breve{y}_j) - b_i(\breve{y}_i - \breve{y}_{i,r}), \\ z_{i,m} = \breve{x}_{i,m} - \alpha_{i,m}^c, s_{i,m} = \alpha_{i,m}^c - \alpha_{i,m-1}, \tag{11}$$

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where $m = 2, \ldots, n$, b_i represents whether the followers and leader are connected, $b_i > 0$ if connected, otherwise, $b_i = 0$, the output of first-order filter is represented as $\alpha_{i,m}^c$, $\alpha_{i,m-1}$ is the virtual control law. Then, define the constant θ as follows:

$$\begin{aligned}
\theta_{i,1} &= \bar{\rho}_{i,2}\|\beta_{i,1}\|^2/((e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}),\\
\theta_{i,m} &= \bar{\rho}_{i,m+1}\|\beta_{i,m}\|^2/(\underline{\rho}_{i,m}\underline{g}_{i,m}), m = 2, \ldots, n-1,\\
\theta_{i,n} &= \|\beta_{i,n}\|^2/(\underline{\rho}_{i,n}\underline{g}_{i,n}),
\end{aligned} \tag{12}$$

where $e_i = \sum_{j=1}^{N_i} a_{i,j}$ and $\underline{g}_{i,l}$ are the unknown constants, $l = 1, \ldots, n$. The estimated error is defined as $\tilde{\theta} = \theta - \hat{\theta}$, where $\hat{\theta}$ denotes the estimation of θ. According to Assumption 1, it is assumed that $\underline{\rho}_{i,l} \le |\rho_{i,l}(t)| \le \bar{\rho}_{i,l}$, $|\rho_{i,l}(t)| \le \bar{\bar{\rho}}_{i,l}$ with unknown positive constants $\underline{\rho}_{i,l}$, $\bar{\rho}_{i,l}$, and $\bar{\bar{\rho}}_{i,l}$, $l = 1, \ldots, n$.

Step 1: Select a Lyapunov function candidate as

$$V_{i,1} = z_{i,1}^2/2 + \underline{\rho}_{i,1}\underline{g}_{i,1}(e_i+b_i)\tilde{\theta}_{i,1}^2/(2\bar{\rho}_{i,2}). \tag{13}$$

From (1) and (11), differentiating $V_{i,1}$ yields

$$\begin{aligned}
\dot{V}_{i,1} = z_{i,1}\{&(e_i+b_i)[\dot{\rho}_{i,1}x_{i,1} + \rho_{i,1}(g_{i,1}(\breve{x}_{i,1}) + d_{i,1}^*)(z_{i,2}\\
&+ s_{i,2} + \alpha_{i,1})/\rho_{i,2} + \rho_{i,1}(f_{i,1}(\breve{x}_{i,1}) + d_{i,1} + \varpi_{i,1})]\\
&- \sum_{m=1}^{N_i} a_{i,m}[\dot{\rho}_{m,1}x_{m,1} + (g_{m,1}(\breve{x}_{m,1}) + d_{m,1}^*)\breve{x}_{m,2}\\
&* \rho_{m,1}\rho_{m,2}^{-1} + \rho_{m,1}(f_{m,1}(\breve{x}_{m,1}) + d_{m,1} + \varpi_{m,1})] - b_i\dot{\rho}_{i,1}y_r\\
&- b_i\rho_{i,1}\dot{y}_r\} - (e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}\tilde{\theta}_{i,1}\dot{\hat{\theta}}_{i,1}/\bar{\rho}_{i,2},
\end{aligned} \tag{14}$$

where $f_{i,1}(\breve{\cdot})$, $g_{i,1}(\breve{\cdot})$ are the post-attack system functions of the ith agent and its neighbor, $d_{\cdot,1}$ is the error between $f_{i,1}(\cdot)$ and $f_{i,1}(\breve{\cdot})$, $d_{\cdot,1}^*$ is the error between $g_{i,1}(\cdot)$ and $g_{i,1}(\breve{\cdot})$, respectively. It is assumed that $d_{\cdot,1}$ and $d_{\cdot,1}^*$ are unknown and satisfy $0 < \underline{\chi}_{\cdot,1} < |d_{\cdot,1}| < \bar{\chi}_{\cdot,1}$ with unknown constants $\underline{\chi}_{\cdot,1}$ and $\bar{\chi}_{\cdot,1}$, $0 < \underline{\chi}_{\cdot,1}^* < |d_{\cdot,1}^*| < \bar{\chi}_{\cdot,1}^*$ with unknown positive constants $\underline{\chi}_{\cdot,1}^*$ and $\bar{\chi}_{\cdot,1}^*$.

From the completion of squares, the following inequalities holds:

$$\begin{aligned}
&z_{i,1}(e_i+b_i)\{\dot{\rho}_{i,1}\breve{x}_{i,1}/\rho_{i,1} + \rho_{i,1}(f_{i,1}(\breve{x}_{i,1}) + d_{i,1} + \varpi_{i,1})\}\\
&\quad - z_{i,1}b_i(\dot{\rho}_{i,1}y_r + \rho_{i,1}\dot{y}_r) \le (\bar{\bar{\rho}}_{i,1}\breve{x}_{i,1}z_{i,1})^2/(2\underline{\rho}_{i,1}^2) + (e_i+b_i)^2/2\\
&\quad + f_{i,1}^2(\breve{x}_{i,1})z_{i,1}^2/2 + z_{i,1}^2 + b_i\bar{\bar{\rho}}_{i,1}^2y_r^2z_{i,1}^2/2 + b_i\bar{\bar{\rho}}_{i,1}^2f^2(x_r)\\
&\quad *z_{i,1}^2/2 + \omega_{i,1},
\end{aligned} \tag{15}$$

$$\begin{aligned}
&- z_{i,1}\{\sum_{m=1}^{N_i} a_{i,m}\dot{\rho}_{m,1}\breve{x}_{m,1}/\rho_{m,1} + \sum_{m=1}^{N_i} a_{i,m}\rho_{m,1}(g_{m,1}(\breve{x}_{m,1})\\
&+ (d_{m,1}^*)\breve{x}_{m,2}/\rho_{m,2}\} \le \sum_{m=1}^{N_i} H_{i,m}(a_{i,m}\bar{\bar{\rho}}_{m,1}\breve{x}_{m,1}z_{i,1})^2/2\underline{\rho}_{m,1}^2\\
&+ \sum_{m=1}^{N_i} M_{i,m}(a_{i,m}\bar{\rho}_{m,1}\bar{g}_{m,1}\breve{x}_{m,2}z_{i,1})^2/(2\underline{\rho}_{m,2}^2) + \omega_{i,2},
\end{aligned} \tag{16}$$

$$\begin{aligned}
&- z_{i,1}\sum_{m=1}^{N_i} a_{i,m}\rho_{m,1}(f_{m,1}(\breve{x}_{m,1}) + d_{m,1} + \varpi_{m,1}) \le \sum_{m=1}^{N_i}\\
&(a_{i,m}\bar{\rho}_{m,1}f_{m,1}(\breve{x}_{m,1})z_{i,1})^2/2 + \sum_{m=1}^{N_i}(a_{i,m}\bar{\rho}_{m,1}\bar{\chi}_{m,1})^2/2\\
&+ \sum_{m=1}^{N_i}(a_{i,m}\bar{\rho}_{m,1}\bar{\varpi}_{m,1})^2/2 + N_i/2 + N_iz_{i,1}^2,
\end{aligned} \tag{17}$$

where $\omega_{i,1} = (e_i+b_i)^2(1 + \bar{\chi}_{i,1}^2 + \bar{\varpi}_{i,1}^2)/2 + b_i$, $\omega_{i,2} = \sum_{m=1}^{N_i} 1/(2H_{i,m}) + 1/(2M_{i,m})$, $\bar{g}_{m,1} = \bar{g}_{m,1}^* + \bar{\chi}_{m,1}^*$, $H_{i,m}$ and $M_{i,m}$ are the positive design parameters. Combining the above inequalities and (14), $\alpha_{i,1}$ is constructed as

$$\alpha_{i,1} = -z_{i,1}\hat{\alpha}_{i,1}/\sqrt{z_{i,1}^2\hat{\alpha}_{i,1}^2 + \varepsilon_{i,1}^2}, \tag{18}$$

where $\varepsilon_{i,1}$ is a positive constant. Next, the following inequality can be verified:

$$\begin{aligned}
&\rho_{i,1}(g_{i,1}(\breve{x}_{i,1}) + d_{i,1}^*)(e_i+b_i)z_{i,1}\alpha_{i,1} \le \bar{\rho}_{i,1}\bar{g}_{i,1}(e_i+b_i)\varepsilon_{i,1}\\
&/\underline{\rho}_{i,2} - \underline{\rho}_{i,1}\underline{g}_{i,1}(e_i+b_i)z_{i,1}\hat{\alpha}_{i,1}/\bar{\rho}_{i,2}.
\end{aligned} \tag{19}$$

where $\bar{g}_{i,1} = \bar{g}_{i,1}^* + \bar{\chi}_{i,1}^*$ and $\underline{g}_{i,1} = \underline{g}_{i,1}^* + \underline{\chi}_{i,1}^*$. According to (19), the following inequality can be obtained:

$$\begin{aligned}
\dot{V}_{i,1} \le z_{i,1}\{&-\underline{\rho}_{i,1}\underline{g}_{i,1}(e_i+b_i)z_{i,1}\hat{\alpha}_{i,1}/\bar{\rho}_{i,2} + F(\sigma_{i,1})\\
&+ (N_i+1)z_{i,1}\} + D_{i,1} - \bar{\rho}_{i,2}\tilde{\theta}_{i,1}\dot{\hat{\theta}}_{i,1}/(2(e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1})\\
&+ z_{i,1}(e_i+b_i)\rho_{i,1}g_{i,1}(x_{i,1})(z_{i,2}+s_{i,2})/\rho_{i,2},
\end{aligned} \tag{20}$$

where $F(\sigma_{i,1}) = \bar{\bar{\rho}}_{i,1}^2\breve{x}_{i,1}^2z_{i,1}/(2\underline{\rho}_{i,1}^2) + f_{i,1}^2(\breve{x}_{i,1})z_{i,1}/2 + \sum_{m=1}^{N_i} H_{i,m}(a_{i,m}\bar{\bar{\rho}}_{m,1}\breve{x}_{m,1})^2z_{i,1}/(2\underline{\rho}_{m,1}^2) + \sum_{m=1}^{N_i} M_{i,m}(a_{i,m}\bar{\rho}_{m,1}\bar{g}_{i,1}\breve{x}\underline{\rho}_{m,2})^2z_{i,1}/(2\underline{\rho}_{m,2}^2)$, $D_{i,1} = (1 + \bar{d}_{i,1}^2/2 + \bar{\varpi}_{i,1}^2/2)(e_i+b_i)^2 + \sum_{m=1}^{N_i}[1/(2H_{i,m}) + 1/(2M_{i,m})] + N_i/2 + b_i + \bar{\rho}_{i,1}\bar{g}_{i,1}*(e_i + b_i)\varepsilon_{i,1}/\underline{\rho}_{i,2} + \sum_{m=1}^{N_i}(a_{i,m}\bar{\rho}_{m,1}f_{m,1}(\breve{x}_{m,1}))^2z_{i,1}/2 + b_i\bar{\bar{\rho}}_{i,1}^2\breve{y}_r^2z_{i,1}/(2\underline{\rho}_{i,1}^2) + b_i(\bar{\rho}_{i,1}(f(\breve{x}_r) + \bar{\chi}_r))^2z_{i,1}/2$. Using the neural networks approximation technology [14], $F(\sigma_{i,1}) = \beta_{i,1}^TS_{i,1}(\sigma_{i,1}) + \tau_{i,1}$, where $\tau_{i,1}$ is the compensation error and $|\tau_{i,1}| \le \bar{\tau}_{i,1}$.

Further, using the completion of squares, one has

$$\begin{aligned}
&z_{i,1}\beta_{i,1}S_{i,1} \le W_{i,1}(e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}\theta_{i,1}z_{i,1}^2q_{i,1}/(2\bar{\rho}_{i,2})\\
&\quad + 1/(2W_{i,1}),
\end{aligned} \tag{21}$$

$$\begin{aligned}
&z_{i,1}(e_i+b_i)\rho_{i,1}g_{i,1}(x_{i,1})(z_{i,2}+s_{i,2})/\rho_{i,2} \le [(e_i+b_i)\bar{\rho}_{i,1}\bar{g}_{i,1}\\
&\quad /\underline{\rho}_{i,2}]^2z_{i,1}^2 + z_{i,2}^2/2 + s_{i,2}^2/2,
\end{aligned} \tag{22}$$

$$z_{i,1}\tau_{i,1} \le U_{i,1}z_{i,1}^2/2 + \tau_{i,1}^2/(2U_{i,1}). \tag{23}$$

Next, select $\hat{\alpha}_{i,1}$ and $\dot{\hat{\theta}}_{i,1}$ as

$$\hat{\alpha}_{i,1} = C_{i,1}z_{i,1}/2 + \bar{C}_{i,1}2^{-\frac{3}{4}}z_{i,1}^{\frac{1}{2}} + W_{i,1}z_{i,1}\hat{\theta}_{i,1}q_{i,1}/2, \tag{24}$$

$$\dot{\hat{\theta}}_{i,1} = W_{i,1}z_{i,1}^2q_{i,1}/2 - R_{i,1}\hat{\theta}_{i,1}, \tag{25}$$

where $C_{i,1}$, $\bar{C}_{i,1}$ and $R_{i,1}$ are the design parameters.

Combining (21)-(25), we can obtain

$$\begin{aligned}
\dot{V}_{i,1} \le &-C_{i,1}^*z_{i,1}^2/2 - \bar{C}_{i,1}^*2^{-\frac{3}{4}}z_{i,1}^{\frac{3}{2}} - R_{i,1}(e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}\tilde{\theta}_{i,1}^2\\
&/(r_{i,1}\bar{\rho}_{i,2}) + \bar{D}_{i,1} + z_{i,2}^2/2 + s_{i,2}^2/2,
\end{aligned} \tag{26}$$

where $C_{i,1}^* = C_{i,1}(e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}/\bar{\rho}_{i,2} - N_i - U_{i,1}/2 - 1 - [(e_i+b_i)\bar{\rho}_{i,1}\bar{g}_{i,1}/\underline{\rho}_{i,2}]^2$, $\bar{C}_{i,1}^* = \bar{C}_{i,1}(e_i+b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}/\bar{\rho}_{i,2}$, $\bar{D}_{i,1} = D_{i,1} + 1/(2W_{i,1}) + \bar{\tau}_{i,1}^2/(2U_{i,1}) + R_{i,1}(e_i + b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}\theta_{i,1}^2/(r_{i,1}\bar{\rho}_{i,2})$.

Step m (m = 2,..., n): Similar to step 1, select a Lyapunov function candidate as

$$V_{i,m} = z_{i,m}^2/2 + \underline{\rho}_{i,m}\underline{g}_{i,m}\tilde{\theta}_{i,m}^2/(2\bar{\rho}_{i,m+1}), \tag{27}$$

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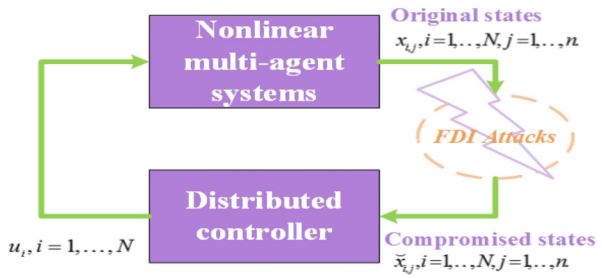


Fig. 1. Design framework of the researched systems (1) with the FDI attacks.

where $\underline{g}_{i,m} = \underline{g}^*_{i,m} + \underline{\chi}^*_{i,m}$. Then, $\alpha_{i,m}$ and $\dot{\hat{\theta}}_{i,m}$ are designed as

$$\alpha_{i,m} = -z_{i,m}\hat{\alpha}_{i,m}/\sqrt{z^2_{i,m}\hat{\alpha}^2_{i,m} + \varepsilon^2_{i,m}}, \tag{28}$$

$$\dot{\hat{\theta}}_{i,m} = W_{i,m}z^2_{i,m}q_{i,m} - R_{i,m}\hat{\theta}_{i,m}, \tag{29}$$

where $\varepsilon_{i,m}$ is a positive constant, $\hat{\alpha}_{i,m} = C_{i,m}z_{i,m}/2 + \bar{C}_{i,m}2^{-\frac{3}{4}}z^{\frac{1}{2}}_{i,m} + W_{i,m}z_{i,m}\hat{\theta}_{i,m}q_{i,m}/2$.

Combining (28) and (29), one has

$$\begin{aligned}\dot{V}_{i,m} \le& -C^*_{i,m}z^2_{i,1}/2 - \bar{C}^*_{i,m}2^{-\frac{3}{4}}z^{\frac{3}{2}}_{i,m} - R_{i,m}\underline{\rho}_{i,m}\underline{g}_{i,m}\tilde{\theta}^2_{i,m}/\\ &(2r_{i,m}\bar{\rho}_{i,m+1}) + \bar{D}_{i,m} + P_{i,m}z^2_{i,m+1}/2\\ &+ P_{i,m}s^2_{i,m+1}/2,\end{aligned} \tag{30}$$

where $C^*_{i,m} = C_{i,m}\underline{\rho}_{i,m}\underline{g}_{i,m}/\bar{\rho}_{i,m+1} - U_{i,m}/2 - 1 + P_{i,m}\bar{\rho}_{i,m}\bar{g}_{i,m}/\underline{\rho}_{i,m+1}$, $\bar{C}^*_{i,m} = \bar{C}_{i,m}\underline{\rho}_{i,m}\underline{g}_{i,m}/\bar{\rho}_{i,m+1}$, $\bar{D}_{i,m} = \bar{\rho}_{i,m}\bar{g}_{i,m}\varepsilon_{i,m}/\underline{\rho}_{i,m+1} + 1/(2W_{i,1}) + \bar{\tau}^2_{i,1}/(2U_{i,1}) + 1 + 1/\bar{d}^2_{i,m} + \bar{\varpi}^{-2}_{i,m}R_{i,m}\underline{\rho}_{i,m}\underline{g}_{i,m}\theta^2_{i,m}/(r_{i,m}\bar{\rho}_{i,m+1})$. It is worth noting that when $m = n$, we have $\bar{\rho}_{m+1} = 1$, $\underline{\rho}_{m+1} = 1$, and $P_{i,m} = 0$.

Furthermore, the design framework of the researched systems (1) with the FDI attacks is shown in Fig. 1. The following theorem are generalized to illustrate the main results of this brief.

Theorem 1: Construct the control laws (18), (28), the adaptive laws (25), (29) for the studied system (1) under Assumption 1-3, Lemma 1-3, and FDI attacks. It is ensured that the studied system is stable and the distributed consensus error converges to a tunable zero neighborhood within the finite-time period.

Proof: Select the whole Lyapunov function V as

$$V = \sum_{i=1}^{N}\left\{\sum_{\kappa=1}^{n}V_{i,\kappa} + \sum_{\kappa=1}^{n-1}s^2_{i,\kappa+1}\right\}. \tag{31}$$

The virtual control $\alpha_{i,\kappa}$ get pass the mth first-order filter with the design parameter $\vartheta_{i,\kappa+1}$ to obtain $\alpha^c_{i,\kappa}$, in which

$$\vartheta_{i,\kappa+1}\dot{\alpha}^c_{i,\kappa+1} + \alpha^c_{i,\kappa+1} = \alpha_{i,\kappa}, \alpha^c_{i,\kappa+1}(0) = \alpha_{i,\kappa}(0). \tag{32}$$

Further, according to (11), one has

$$\dot{s}_{i,\kappa+1} = -s_{i,\kappa+1}/\vartheta_{i,\kappa+1} - \dot{\alpha}_{i,\kappa}. \tag{33}$$

It is assumed that $|-\dot{\alpha}_{i,\kappa}| \le \zeta_{i,\kappa+1}(z_{i,\kappa}, y_r, \dot{y}_r, \ddot{y}_r, \bar{\hat{\theta}}_{i,\kappa})$, where the continuous function $\zeta_{i,\kappa+1}(z_{i,\kappa}, y_r, \dot{y}_r, \ddot{y}_r, \bar{\hat{\theta}}_{i,\kappa})$ satisfies $\zeta_{i,\kappa+1} < \bar{\zeta}_{i,\kappa+1}$ and $\bar{\hat{\theta}}_{i,\kappa} = [\hat{\theta}_{i,1}, \ldots \hat{\theta}_{i,\kappa}]$. By combining (32) and (33) with the completion of squares and using Lemma 3, for $m_1 = 1 - m_2$, $m_2 = 3/4$, $m_3 = m_1m^3_2$, $\lambda_1 = 1$, $\lambda_2 = s^2_{i,\kappa}/2$, $\lambda_3 = 1$, one has

$$s_{i,\kappa+1}\dot{s}_{i,\kappa+1} \le (\vartheta_{i,\kappa} - 2)s^2_{i,\kappa+1}/2\vartheta_{i,\kappa+1} + \bar{\zeta}^2_{i,\kappa+1}/2, \tag{34}$$

$$(s^2_{i,\kappa+1}/2)^{\frac{3}{4}} \le s^2_{i,\kappa+1}/2 + \kappa_3. \tag{35}$$

Subsequently, using Lemma 2 and Lemma 3, for $m_1 = 1 - m_2$, $m_2 = 3/4$, $m_3 = m_1m^3_2$, $\lambda_1 = 1$, $\lambda_2 = \sum_{\kappa=1}^{n}R_{i,\kappa}\underline{\rho}_{i,\kappa}\underline{g}_{i,\kappa}\tilde{\theta}^2_{i,\kappa}/(r_{i,\kappa}\bar{\rho}_{i,\kappa+1})$, $\lambda_3 = 1$, one has

$$\begin{aligned}\dot{V} \le& \sum_{i=1}^{N} -\gamma_i[\sum_{\kappa=1}^{n}z^2_{i,\kappa}/2] - \bar{\gamma}_i[\sum_{\kappa=1}^{n}z^2_{i,\kappa}/2]^{\frac{3}{4}} - (\gamma^*_i - 1)[(e_i + b_i)\\ &\underline{\rho}_{i,1}\underline{g}_{i,1}\tilde{\theta}^2_{i,1}/(2r_{i,1}\bar{\rho}_{i,2}) + \sum_{\kappa=1}^{n-1}\underline{\rho}_{i,\kappa}\underline{g}_{i,\kappa}\tilde{\theta}^2_{i,\kappa}/(2r_{i,\kappa}\\ &\bar{\rho}_{i,\kappa+1}) + \underline{\rho}_{i,n}\underline{g}_{i,n}\tilde{\theta}^2_{i,n}/(2r_{i,n})] - [(e_i + b_i)\underline{\rho}_{i,1}\underline{g}_{i,1}\tilde{\theta}^2_{i,1}/(2r_{i,1}\\ &\bar{\rho}_{i,2})]^{\frac{3}{4}} - [\sum_{\kappa=1}^{n-1}\underline{\rho}_{i,\kappa}\underline{g}_{i,\kappa}\tilde{\theta}^2_{i,\kappa}/(2r_{i,\kappa}\bar{\rho}_{i,\kappa+1})]^{\frac{3}{4}} - [\underline{\rho}_{i,n}\underline{g}_{i,n}\tilde{\theta}^2_{i,n}\\ &/(2r_{i,n})]^{\frac{3}{4}} - \widehat{\gamma}_i\sum_{\kappa=1}^{n-1}s^2_{i,\kappa+1} - \sum_{\kappa=1}^{n-1}(s^2_{i,\kappa+1}/2)^{\frac{3}{4}} + D^*\\ \le& -\gamma V - \bar{\gamma}V^{\frac{3}{4}} + D^*,\end{aligned} \tag{36}$$

where $\gamma_{i,1} = \min\{C^*_{i,1}, \ldots, C^*_{i,n}\}$, $\bar{\gamma}_{i,1} = \min\{\bar{C}^*_{i,1}, \ldots, \bar{C}^*_{i,n}\}$, $\gamma^*_{i,1} = \min\{R_{i,1}, \ldots, R_{i,n}\}$, $\widehat{\gamma}_i = \min\{(-\vartheta_{i,2} - 2)/(2\vartheta_{i,3}), \ldots, (-\vartheta_{i,n-1} - 2)/(2\vartheta_{i,n})\}$, $\gamma = \min\{\gamma_i, (\gamma^*_i - 1), \gamma_i, \widehat{\gamma}_i\}$, $\bar{\gamma} = \min\{\bar{\gamma}_i, 1\}$, and $D^* = \sum_{\kappa=1}^{n-1}\bar{D}_{i,\kappa} + (2n-1)m_3$.

Furthermore, (36) can be reconstructed as follows:

$$\dot{V} \le -(1-\mu)\gamma V - \mu\gamma V - \bar{\gamma}V^{\frac{3}{4}}, \tag{37}$$

where $0 < \mu < 1$. According to (31) and Lemma 1, one has

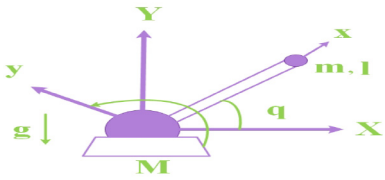
$$\lim_{t\to T} V \le \min\{\frac{D^*}{(1-\mu)\gamma}, [\frac{D^*}{(1-\mu)\bar{\gamma}}]^{\frac{3}{4}}\}. \tag{38}$$

Therefore, the practical finite time with the setting time T can be obtained as $T \le \max\{t_0 + 4\ln\frac{\mu\gamma V^{\frac{1}{4}}(t_0)+\bar{\gamma}}{b}/\mu\gamma, t_0 + 4\ln\frac{\gamma V^{\frac{1}{4}}(t_0)+\mu\bar{\gamma}}{\mu\bar{\gamma}}/\gamma\}$. Moreover, due to the boundedness of V, the boundedness of the errors $z_{i,\kappa}$, $\tilde{\theta}_{i,\kappa}$, $s_{i,\kappa}$ is ensured. According to the boundedness of $\tilde{\theta}_{i,\kappa}$ and $\theta_{i,\kappa}$, it can be obtained that $\hat{\theta}_{i,\kappa}$ is bounded. Further, it is deduced that $\alpha_{i,\kappa}$ is bounded. Next, according to (11), the boundedness of $\alpha^c_{i,\kappa}$ can be obtained. According to the definition of $z_{i,1}$ and (38), one has

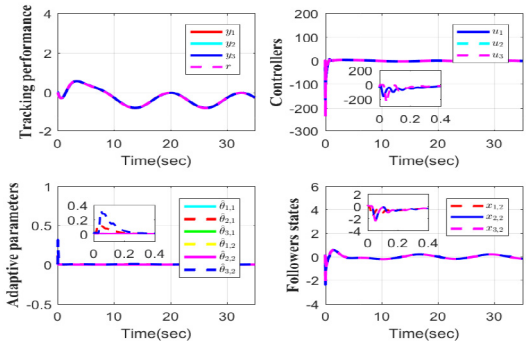
$$\begin{aligned}\|y - y_r\| \le& \min\{\sqrt{D^*/(1-\mu)\gamma},\\ &\sqrt{[D^*/(1-\mu)\bar{\gamma}]^{\frac{3}{4}}}\}/[\sigma_1\underline{\varsigma}(L+B)]\},\end{aligned} \tag{39}$$

where y is the column vector of output of all followers, $y_r = (y_r, \ldots, y_r)^T$, $\underline{\varsigma}(L+B)$ is the least singular value and $\sigma_1 = \min\{\underline{\rho}_{1,1}, \underline{\rho}_{2,1}, \ldots, \underline{\rho}_{N,1}\}$. In the end, the boundedness of all closed-loop signals is ensured. ■

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(a) Single-link robot model.



(b) The trajectories of all closed-loop signals.

Fig. 2. System model and the trajectories of all closed-loop signals.

IV. Simulation Study

To prove the applicability of the proposed security control strategy, the single-link robot is considered (see Fig. 2 (a)). Then, the ith single-link robot is modeled:

$$M\ddot{q}_i + mgl\sin(q_i) = u_i, y_i = q_i, i = 1, 2, 3, \tag{40}$$

where the moment of inertia at the junction is given as $M = 0.5$, q_i denotes the angle between the manipulator and the horizontal platform, the acceleration of gravity is given as $g = 9.8$, the mass of the connecting rod is given as $m = 1$, the length of the connecting rod is given $l = 1$, the input torque is expressed as u_i. When $t \geq 2$, $\rho_{i,1}$ is chosen as $\rho_{i,1} = 0.05e^{7-t} + 0.95$. When $t \geq 1$, $\rho_{i,2}$ is chosen as $\rho_{i,2} = 0.3\sin(t) + 0.7$. The initial states of the system (40) are chosen as $[0, 0.2]$. The model of leader is described as $\dot{x}_r = -1.2x_r + 0.5\cos(t/2 + 3) + 3t^2e^{-t} - 0.5$, $y_r = x_r$.

The adjacency matrix A and the Laplacian matrix L of the system (39) are defined as $A = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, $L = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$, respectively. The simulation results are described in Fig. 2 (b), which show the output trajectories of all agents y_i and the leader y_r, the trajectories of control signal of each follower, the trajectories of the adaptive laws $\hat{\theta}_{i,j}$, the trajectories of the system state $x_{i,2}$ of each follower, respectively. Therefore, it is demonstrated in all these simulation results that the proposed security control strategy can obtain desired performance.

V. Conclusion

In this brief, the distributed finite-time consensus tracking control problem for the nonlinear multi-agent systems with the FDI attacks was investigated. By improving the coordinate transformation, the more practical situations are considered in this brief, where different system states of the one follower suffer from different attack signals. The constructed security algorithm which guarantees the outputs of all agents can achieve consensus and that all the signals of the studied system are bounded within the finite-time period. The security control problem for cyber-physical systems under replay attacks will be considered in the future.

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